

CZESTOCHOWA, Poland

WMO: 125500

Lat: 50.8118N

Lon: 19.0915E

Elev: 295

StdP: 97.83

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.1	-16.9	0.9	-12.8	-14.6	1.1	-10.0	8.3	1.6	7.2	1.2	2.2	90	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	30.1	19.6	28.1	18.7	26.5	18.0	20.7	27.5	19.9	26.1	19.1	24.6	2.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.9	23.0	17.7	13.2	22.2	17.0	12.5	21.4	60.9	28.0	58.1	26.3	55.4	24.8	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.7	5.1	DB	-16.5	33.2	4.4	1.6	-19.7	34.4	-22.3	35.4	-24.7	36.3	-27.9	37.5	(4)
(5)			WB	-16.9	22.4	4.3	0.7	-20.0	22.9	-22.5	23.4	-24.9	23.8	-28.0	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-1.7	-0.8	3.3	9.6	13.9	17.6	19.5	18.9	14.4	9.4	4.9	0.4	(6)	
	DBStd	8.56	5.22	5.04	4.73	4.36	4.05	3.62	3.42	3.20	3.49	4.32	4.16	4.41	(7)	
	HDD10.0	1469	363	302	211	58	11	0	0	0	4	66	157	297	(8)	
	HDD18.3	3508	622	535	466	262	147	56	28	32	126	277	402	555	(9)	
	CDD10.0	1168	0	0	3	47	131	229	294	276	136	47	5	0	(10)	
	CDD18.3	166	0	0	0	1	9	34	64	50	7	1	0	0	(11)	
	CDH23.3	1372	0	0	0	16	89	286	526	394	59	1	0	0	(12)	
	CDH26.7	342	0	0	0	2	13	68	148	103	9	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.8	2.8	2.9	2.8	2.5	2.3	2.2	2.0	2.2	2.3	2.6	2.8	(14)	
(15) Precipitation	PrecAvg	650	38	34	42	43	74	78	95	60	60	44	45	36	(15)	
	PrecMax	887	89	76	99	97	208	176	250	124	148	93	82	110	(16)	
	PrecMin	452	7	7	15	2	29	26	39	22	12	4	0	11	(17)	
	PrecStd	102	17	18	21	20	37	35	49	30	30	27	18	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	12.4	18.2	25.6	28.4	31.5	32.5	32.1	27.9	22.7	16.5	11.9	(19)	
		MCWB	7.3	7.5	10.2	14.8	17.3	20.4	20.5	20.4	17.1	15.4	11.7	8.6	(20)	
	2%	DB	7.9	9.7	15.1	22.0	25.9	28.6	30.3	29.2	25.0	20.3	14.2	9.1	(21)	
		MCWB	6.0	6.2	8.7	12.5	16.5	19.1	19.6	19.4	17.1	14.5	10.5	7.2	(22)	
	5%	DB	6.3	7.7	12.7	19.4	23.6	26.4	28.2	27.3	22.8	18.0	12.2	7.4	(23)	
		MCWB	4.7	5.0	7.6	11.4	15.6	17.9	18.9	18.8	15.9	13.7	9.7	5.9	(24)	
	10%	DB	4.8	5.6	10.4	17.3	21.4	24.6	26.3	25.4	20.9	15.9	10.5	5.9	(25)	
		MCWB	3.6	3.7	6.7	10.4	14.5	17.0	18.1	18.0	15.0	12.6	8.7	4.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	8.3	10.9	15.0	18.6	21.5	22.1	21.5	18.7	16.4	12.3	9.1	(27)	
		MCDB	9.1	11.3	16.5	24.4	26.0	28.8	29.8	28.3	25.4	20.8	15.1	11.2	(28)	
	2%	WB	6.2	6.6	9.4	13.6	17.3	20.1	20.8	20.4	17.6	15.1	11.2	7.3	(29)	
		MCDB	7.6	9.1	13.9	20.4	23.8	26.5	27.4	27.4	23.0	19.0	13.6	8.9	(30)	
	5%	WB	4.8	5.2	8.2	12.3	16.3	19.0	20.0	19.6	16.6	14.0	9.9	6.1	(31)	
		MCDB	6.1	7.4	11.8	18.1	22.0	24.9	26.0	25.8	21.4	17.4	11.9	7.2	(32)	
	10%	WB	3.6	3.9	7.0	11.0	15.2	18.0	19.2	18.7	15.6	12.8	8.7	4.9	(33)	
		MCDB	4.7	5.4	10.1	16.3	20.6	23.1	24.5	24.0	19.9	15.7	10.3	5.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.6	7.4	9.5	9.4	9.4	9.6	9.6	8.9	7.6	5.4	4.4	(35)	
		MCDBR	5.4	8.4	11.5	13.0	12.8	12.4	12.3	12.5	12.0	10.6	8.3	5.7	(36)	
	5% WB	MCWBR	4.2	5.4	6.6	6.1	5.3	5.1	4.2	4.8	5.3	5.7	5.3	4.4	(37)	
		MCDWR	5.2	7.5	10.4	11.8	11.6	11.2	10.8	11.1	10.6	10.0	7.5	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.316	0.341	0.380	0.432	0.426	0.422	0.445	0.434	0.401	0.370	0.347	0.313	(40)	
		taud	2.275	2.262	2.214	2.132	2.202	2.244	2.192	2.225	2.287	2.340	2.342	2.327	(41)	
	Edn at Noon		692	775	809	805	831	837	808	796	780	733	651	639	(42)	
		Edh at Noon	69	91	115	140	136	132	137	126	106	84	66	57	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.79	1.52	2.67	4.09	5.04	5.30	5.04	4.51	3.23	1.89	0.91	0.61	(44)	
	RadStd		0.15	0.27	0.37	0.53	0.58	0.53	0.62	0.37	0.45	0.27	0.12	0.08	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+0.98	+0.78	N/A	N/A	N/A	N/A	(46)
(47)	Regional (34 neighbors)	+0.50	N/A	N/A	+0.78	+0.33	N/A	N/A	-164	+67	+26	(47)

Nomenclature: See separate page